

MASCOT
GOES
HERE

KID INSTRUCTIONS

Last time, you learned to spot loops everywhere — in songs, dances, even your weekly chores. Now level up: it's your turn to write loops of your own!

Repeat

Repeat — To do the same thing again and again. In a loop, "repeat" tells the computer how many times to do something.

Example: "Repeat 'do a push-up' 10 times" is a loop with 10 push-ups inside it.

Basic Version

Loop It! — Turn each task into a loop: **Repeat "[action]" [N] times.**

Example: Brushing each tooth (32 teeth) → **Repeat "brush one tooth" 32 times.**

Scenario 1

Doing 20 push-ups

→

Scenario 2

Watering 6 plants on the windowsill (one cup of water each)

→

Scenario 3

High-fiving every kid in a line of 8 friends

→

Scenario 4

Putting one sticker on each of 12 birthday cards

→

Challenge Version

Loops with a Twist — Sometimes you repeat **until** something happens, or put a loop **inside** another loop!

PART A — Loops with a CONDITION

You don't always know how many times. Repeat **until** a condition is true. Format: **Repeat "[action]" until [condition].**

Example: Walking up the stairs → **Repeat "take one step up" until you reach the top.**

Scenario A1

Pouring water into a glass (you stop when it's full)

Scenario A2

Practicing free throws (you stop when you make 10 in a row)

PART B — NESTED LOOPS

A loop **INSIDE** another loop. Like 3 sets of 10 push-ups — the outer loop is "do 3 sets," and inside each set is "do 10 push-ups."

Repeat 3 times:

Repeat "do 1 push-up" 10 times.

Rest for 30 seconds.

Scenario B

A teacher hands out a quiz to 4 rows of desks. Each row has 5 students. The teacher walks down each row and gives one quiz to each student.

Hint: outer loop = 4 rows. Inner loop = 5 students per row.

Self-Check

- ☐ I wrote a loop for all 4 Basic scenarios using the correct format
- ☐ I wrote 2 condition-based loops using "Repeat ... until ..."
- ☐ I wrote 1 nested loop with an outer and inner loop
- ☐ I can explain in my own words what "nested" means

Reflection Box

Look at one of the loops you wrote. If the number changed — like "32 teeth" became "320 teeth" — would your loop still work the same way? Why is that powerful?

Bonus: Design a Loop for a Robot

You're programming a small cleaning robot for your bedroom! The robot can only do the actions in its list — nothing else.

Robot's Available Actions

- pick up toy
- put toy in bin
- vacuum floor square
- move forward
- turn left
- turn right

Mission: Your bedroom has 8 toys on the floor that need to go in the toy bin, AND a 3×3 grid of carpet squares (9 squares total) that need to be vacuumed. Write a loop (or loops!) to tell the robot how to clean the room. You can use as many loops as you need.

Bonus: Did you use a nested loop? Why or why not?

TEACHER / PARENT NOTE

This activity moves kids from recognizing loops (Activity 3) to writing them. The Challenge version introduces two loop variants they'll meet again in real code: condition-based loops (while-loops, which run until something becomes true) and nested loops (loops inside loops, used in everything from grids to sorting). Discussion prompt: Ask your child to spot one place in their home or school day where a nested loop happens — for example, a morning routine repeated across 5 weekdays is a nested loop.